

The first term in Eq. (2.51) is the data-fitting cost. The second term in Eq. (2.51) is the regularization cost, which enforces the logarithm of background irradiance to zero (i.e., the background irradiance is driven to be a constant value, 1). Further, Ψ is a set of representative background pixel samples. The constant background will facilitate cell image analysis, such as cell image segmentation by simple thresholding. In reality, the irradiance of background pixels may not be a constant due to natural variations. The goal of this constraint is to recover a camera response function similar to Fig. 2.12. Given a wide range of background pixel intensities, their corresponding irradiance values are recovered into a small range around zero. This regularization is still applicable when there is a slow irradiance gradient across the image. If the value range of the background irradiance is different from the cell irradiance, we can still segment cells out of the background. In fact, the irradiance from the background is different from the irradiance from cells in the phase contrast microscope. Otherwise, we cannot visualize cells through the ocular lens using human eyes.

The third term in Eq. (2.51) is a smooth term avoiding overfit on the data, which is defined by the second-order derivative:

$$g''(I) = g(I-1) - 2g(I) + g(I+1) \quad (2.52)$$

The second derivative measures the discontinuity of the curve function. When the second derivative is high, the curve has a sharp curvature. Therefore, minimizing the second derivative can ensure the smoothness during curve fitting. A common camera response function has a steep slope at the end of the camera response curve due to over-saturation or underexposure. In phase contrast microscopy images, the objects of interest (cells) are either dark or bright, while the background pixels have a broad value range in the middle. To visualize the cells better, we can introduce a weight function $\omega(I)$ to

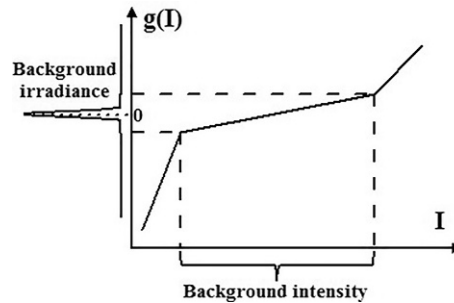


Fig. 2.12 An estimated camera response function such that the logarithm of irradiance values ($g(I)$) of background pixels are recovered to a small range close to zero, while the irradiance details of cell pixels are well recovered by the two ends of the camera response function.

emphasize the smoothness of the two ends of the camera response function. For example, a simple inverted hat function is a good choice for $\omega(I)$:

$$\omega(I) = \left| I - \frac{I_{min} + I_{max}}{2} \right| \quad (2.53)$$

The inverted hat function puts more emphasis on the two ends of the camera response curve (i.e., sensitive to cells) and less in the middle (i.e., to have a flat background region).

In Eq. (2.51), α and β are coefficients that balance the three cost terms. In our experiments, we choose fixed coefficients with $\alpha = 1000$ and $\beta = 1000$.

Because the objective function in Eq. (2.51) is quadratic, taking its derivatives related to g and E and setting them to zero create an overdetermined system of linear equations, ($\mathbf{Ax} = \mathbf{b}$), which can be solved using the pseudo-inverse or singular value decomposition method. The unknown variable in the linear equation system is $\mathbf{x} = [g(I_{min}), \dots, g(I_{max}), -\log E_1, \dots, \log E_N]$, whose first part ($[g(I_{min}), \dots, g(I_{max})]$) is the cell-sensitive camera response function, a lookup table. Given N pixels from P multiexposed images, we have $(I_{max} - I_{min} + 1) + N$ unknown variables in $\mathbf{x}$. To avoid rank deficiency in the linear equation system, we require the number of data-fitting equations $NP > (I_{max} - I_{min} + 1) + N$. For our experiments, $P=6$, $I_{min}=0$, and $I_{max}=4095$. We choose $N=1000$ to satisfy this requirement.

3.3.2 Restoring the irradiance signal

After the cell-sensitive camera response function is estimated from the first set of multiexposed images, for any other sets in the time-lapse image sequence, we can easily restore their relative irradiance signals from image pixel values by

$$\log E_i = \frac{\sum_{j=1}^P w(I_{ij}) (g(I_{ij}) - \Delta t_j)}{\sum_{j=1}^P w(I_{ij})} \quad (2.54)$$

where all the exposures are used to restore the irradiance signal robustly.

Fig. 2.13A shows one of a set of multiexposed phase contrast microscopy images. Fig. 2.13B is the restored irradiance signal map, which clearly demonstrates that the non-cell background region has a uniform background and the contrast between cells and background in the restored irradiance signal map is high. We sort the irradiance values in ascending order (Fig. 2.13C). The majority of the 1.4 million pixels of a 1040×1388 signal map has $E=1$ (i.e., $\log E=0$, which is the goal of the second regularization term in Eq. 2.51). The restored irradiance signal can facilitate the cell image segmentation tasks. For example, by selecting two thresholds T_H and T_L , the microscopy image can be easily thresholded to detect the cells and halo regions (Fig. 2.13D).

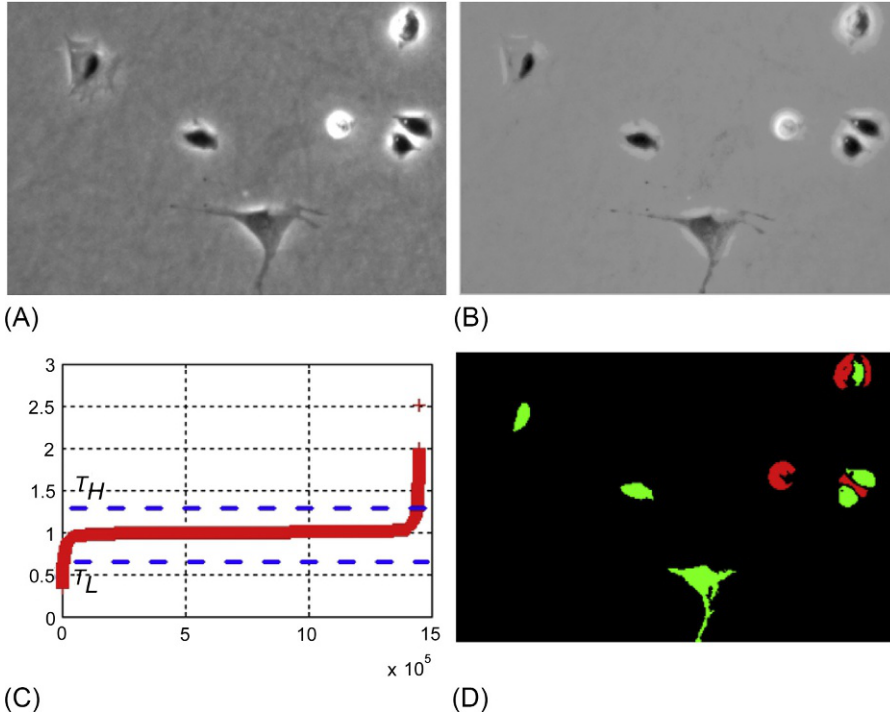


Fig. 2.13 Cell-sensitive imaging. (A) One of a set of multiexposed phase contrast microscopy images; (B) restored irradiance signal map with constant irradiance values on the background and high signal contrast; (C) the sorted irradiance signal values; (D) segmentation by thresholding the irradiance signals.

4. Cell segmentation

The restoration of microscopy images facilitates the subsequential segmentation of cells. In this section, we present various methods to implement cell segmentation, which utilizes a small number of labeled atoms, together with large amounts of unlabeled atoms, to build a better classifier. The scheme requires less human effort and leads to a better result.

4.1 Generation of phase-homogeneous superpixel

Rather than using the pixel grid, we adopt a method to partition a microscopy image into superpixels and utilize them as elements for cell segmentation. Aggregating pixels into superpixels eliminates the local redundancy of an image, which preserves local structures but avoids oversegmented, noisy clips. The computational cost is also significantly reduced without sacrificing the final cell segmentation accuracy, because the partition greatly reduces the number of instances for classification. Moreover, it allows for

measuring an atom's statistical characteristics (e.g., distribution of phase retardation features) rather than a pixel's individual characteristics (e.g., intensity).

Compared with conventional superpixels based on image intensities [1], superpixels in this section are generated by clustering neighboring pixels with their pairwise similarity of phase retardation features. Therefore, phase retardation is statistically homogeneous within a superpixel, which is called a *phase-homogeneous superpixel*. Given two feature vectors at positions $\mathbf{x}_i$ and $\mathbf{x}_j$, the dissimilarity between $\Psi(\mathbf{x}_i)$ and $\Psi(\mathbf{x}_j)$ is defined as

$$d_p = \|\Psi(\mathbf{x}_i) - \Psi(\mathbf{x}_j)\|_2 \leq \|\Psi(\mathbf{x}_i)\|_2 + \|\Psi(\mathbf{x}_j)\|_2 \leq 2\sqrt{K} \quad (2.55)$$

Considering the spatial proximity, the spatial distance between $\mathbf{x}_i$ and $\mathbf{x}_j$, which are expected to be within the same superpixels, is

$$d_s = \|\mathbf{x}_i - \mathbf{x}_j\|_2 \leq S \quad (2.56)$$

where $S = P/N$ denotes the approximate size of each superpixel, with P being the number of image pixels and N being a user-specified number of superpixels.

The partition of superpixels $\{\mathcal{A}_n\}_{n=1}^N$ is determined by minimizing the normalized distortion of phase retardation features and spatial positions as

$$\min \sum_{n=1}^N \sum_{\mathbf{x} \in \mathcal{A}_n} \left(\frac{1}{4K} \|\Psi(\mathbf{x}) - \mu_p^{(n)}\|_2^2 + \frac{\omega_s}{S^2} \|\mathbf{x} - \mu_s^{(n)}\|_2^2 \right) \quad (2.57)$$

where $\mathbf{x}$ is the spatial coordinate; $\Psi(\mathbf{x})$ is the corresponding phase retardation feature; and $\mu_p^{(n)}$ and $\mu_s^{(n)}$ are the prototype and mean of phase retardation features and spatial coordinates for superpixel $\mathcal{A}_n$, respectively. Further, ω_s is a balance variable to adjust the weight between the phase and spatial proximities. The minimization problem can be solved by associating each pixel to its nearest seed iteratively.

Fig. 2.14 shows an example of the atom generation based on phase retardation features, compared with superpixels by image intensities. The phase-homogeneous atoms

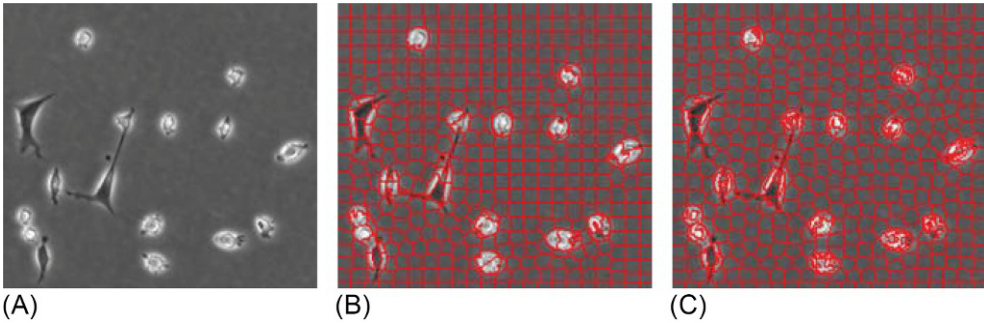


Fig. 2.14 Superpixel generation (best viewed in color). (A) Original phase contrast microscopy image; (B) phase-homogeneous superpixels; (C) intensity-homogeneous superpixels.

preserve the local structure of cell images (i.e., cell regions and halos are segmented into separated atoms, which is consistent with biophysical content of cell images). Nevertheless, the intensity-homogeneous atoms are more likely to produce oversegmented, noisy clips because cell intensities within cells often vary among subcellular structures.

In this case, cell segmentation can be performed by classifying the phase-homogeneous superpixels into several clusters (e.g., bright and dark cells, halos, background). We model the characteristic of feature vectors $\{\Psi(\mathbf{x})\}$ within a specific superpixel $\mathcal{A}_n$, using a multivariate Gaussian distribution with mean vector μ_n and covariance matrix Σ_n , as

$$\mu_n = \frac{1}{\zeta_n} \sum_{i=1}^{\zeta_n} \Psi(\mathbf{x}_i), \quad \Sigma_n = \frac{1}{\zeta_n - 1} \sum_{i=1}^{\zeta_n} (\Psi(\mathbf{x}_i) - \mu_n)(\Psi(\mathbf{x}_i) - \mu_n)^T \quad (2.58)$$

where ζ_n is the number of pixels in superpixel $\mathcal{A}_n$.

In revealing the identity of a superpixel, its neighborhood information plays an important role, in addition to the characteristics of itself. For example, without neighboring information, halos and bright cells may not be distinguishable from each other based on phase retardation features within a superpixel. Specifically, both halos and bright cells are represented by similar distributions of phase retardations. In such a case, neighborhood information helps to distinguish them because most halos are around dark cell regions, while bright cells are surrounded by background.

To explore the relation of a superpixel $\mathcal{A}_n$ with its neighboring superpixel $\mathcal{A}_m$, we define the *neighborhood feature* of $\mathcal{A}_n$ as its most distinctive neighboring superpixel, which is measured with the pairwise Kullback-Leibler divergence to measure the dissimilarity between any pair of superpixels. By combining the feature within a superpixel and the most distinctive one, we obtain the descriptor $\mathbf{f}_n$ for a superpixel $\mathcal{A}_n$.

4.2 Semisupervised cell segmentation

In this section, we introduce a semisupervised classification approach to segment cells out of the background by classifying the partitioned superpixels into specific clusters. Based on a few phase-homogeneous superpixels annotated in each image, identities of the unlabeled superpixels are determined by propagating the labels from the user-specified superpixels. Cell segmentation can be attained based on the classification results.

Mathematically, we define the set of annotated superpixels as $\mathcal{L} \triangleq \{\mathbf{x}_l, \mathbf{y}_l\}_{l=1}^{N_l}$, where N_l is the number of annotated superpixels; $\mathbf{x}_l$ is a feature vector that describes the visual characteristics of the l th labeled superpixel; and $\mathbf{y}_l$ is a binary indicator corresponding to the identity of the l th superpixel—that is, $\mathbf{y}_l(k) = 1$ means that the l th superpixel is labeled as belonging to the k th class. In the same way, we define $\mathcal{U} \triangleq \{\mathbf{x}_u, \mathbf{y}_u\}_{u=1}^{N_u}$, with N_u being the number of unlabeled superpixels and $\{\mathbf{x}_u\}$ being the feature vector of the unlabeled superpixels. The label vectors of unlabeled superpixels $\{\mathbf{y}_u\}$ are going to be inferred via the label propagation procedure.

First, we construct an $\in$ -nearest neighboring graph to characterize the pairwise similarity between superpixels. The adjacency matrix of the weight graph is denoted as $\mathbf{W}=[w_{ij}]$, which is an $N \times N$ nonnegative and symmetric matrix, with $N=N_l+N_u$ being the number of total superpixels. To implement label propagation, the Laplacian matrix $\mathbf{L}$ that corresponds to $\mathbf{W}$ (i.e., $\mathbf{L}=\mathbf{D}-\mathbf{W}$, with $\mathbf{D}$ being the diagonal degree matrix) is rearranged by splitting it into labeled and unlabeled submatrices:

$$\mathbf{L} \leftarrow \begin{bmatrix} \mathbf{L}_{ll} & \mathbf{L}_{lu} \\ \mathbf{L}_{ul} & \mathbf{L}_{uu} \end{bmatrix} \quad (2.59)$$

where $\mathbf{L}_{ll}$ is the Laplacian submatrix that characterizes the relationship between labeled superpixels; $\mathbf{L}_{uu}$ denotes the submatrix corresponding to the unlabeled superpixels; $\mathbf{L}_{lu}$ is a submatrix that interrelates the labeled and unlabeled superpixels; and $\mathbf{L}_{ul}=\mathbf{L}_{lu}^T$. Then, the problem of label propagation is formulated in a matrix form [8] as

$$\mathbf{Y}_u^* = \arg \min_{\mathbf{Y}_u} \text{tr} \left([\mathbf{Y}_l; \mathbf{Y}_u]^T \begin{bmatrix} \mathbf{L}_{ll} & \mathbf{L}_{lu} \\ \mathbf{L}_{ul} & \mathbf{L}_{uu} \end{bmatrix} [\mathbf{Y}_l; \mathbf{Y}_u] \right) \quad (2.60)$$

where tr is the trace operator and $\mathbf{Y}_l$ and $\mathbf{Y}_u$ are binary indicator matrices corresponding to the labeled and unlabeled superpixels, respectively. Specifically, the indicator matrices $\mathbf{Y}_l$ and $\mathbf{Y}_u$ are constructed by stacking up the binary row indicators $\{\mathbf{y}_n\}_{n=1}^N$ in corresponding rows; namely, for the labeled superpixels $\mathbf{Y}_l=[\mathbf{y}_1; \cdots; \mathbf{y}_{N_l}]$, and for the unlabeled superpixels $\mathbf{Y}_u=[\mathbf{y}_{N_l+1}; \cdots; \mathbf{y}_u; \cdots; \mathbf{y}_{N_l+N_u}]$.

Based on the work by Zhu et al. [11], label propagation is solved by Gaussian fields harmonic functions (GFHFs), and the solution is given by a hypothesis $\mathcal{H}$ as

$$\begin{aligned} \mathcal{H}: \mathbf{Y}_u^* &= -(\mathbf{D}_{uu} - \mathbf{W}_{uu})^{-1} \mathbf{L}_{ul} \mathbf{Y}_l \\ &= -\mathbf{L}_{uu}^{-1} \mathbf{L}_{ul} \mathbf{Y}_l \end{aligned} \quad (2.61)$$

where $\mathbf{D}_{uu}$ is a diagonal degree matrix with $\mathbf{D}_{uu}(i, i) = \sum_{j=1}^{N_l+N_u} \mathbf{W}(i, j)$. Note that the elements of $\mathbf{Y}_u^*$ are not binary numbers, but are real; thus, $\mathbf{Y}_u^*$ can be considered as soft-label results. The hard-label vector can be obtained simply by converting the maximum value in each $\mathbf{y}_u^*$ into 1 and the others into 0. Finally, cell segmentation is conducted by grouping the neighboring superpixels with the same labels.

Fig. 2.15 shows some sample results of image segmentation with label propagation, which demonstrates that high-quality cell segmentation is realized with only a small amount of human annotations. We first partition the phase contrast microscopy images into phase-homogeneous superpixels by aggregating the neighboring pixels with similar phase retardation features, as is shown in Fig. 2.15C. Consequentially, the superpixels are utilized as elements for image segmentation. The superpixels marked as *gray*, *light gray*, and *black* are annotated as bright cells, dark cells, and background regions. Fig. 2.15D shows the soft segmentation results based on label propagation, in which the *gray*, *light*

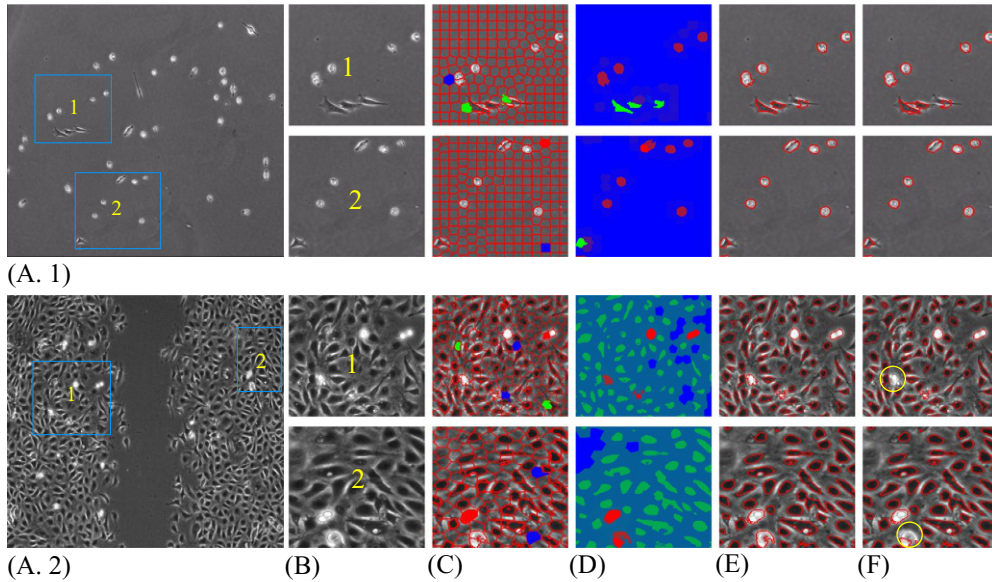


Fig. 2.15 Sample results of cell segmentation. (A) Input phase contrast microscopy images. (B) Zoom-in subimages. (C) Sample selection and annotation over the phase-homogeneous superpixels, where the *gray*, *light gray*, and *black* annotations indicate the corresponding superpixels are annotated as bright cells, dark cells, and background, respectively. (D) Soft classification results based on label propagation with human annotations. The *gray*, *light gray*, and *black* channel values in the soft classification results indicate the likelihoods of the superpixels belonging to bright cell, dark cell, and background regions, respectively. (E) Cell segmentation by finding the labels with the maximum likelihood, and grouping the neighboring superpixels with the same labels. (F) Cell segmentation based on linear imaging models as comparison.

gray, and *black* channel values in the soft segmentation indicate the likelihoods of the superpixels belonging to bright cell, dark cell, and background regions, respectively. Each soft segmentation result was converted into a hard segmentation result by finding the label with the maximum likelihood, and grouping the neighboring superpixels with the same labels, as shown in Fig. 2.15E. For comparison, we also implemented cell segmentation based on the linear imaging model, as is shown in Fig. 2.15F. The results demonstrate that there are false cell detections for some challenging cases, as marked with white circles. Halos are recognized as cell regions in the first row, when cells form as clusters, and some bright cells with low contrast to the background are also missed in the second row.

4.3 Online correction of cell segmentation

Although the previous approach of cell segmentation has been shown to be effective, but it still is not error-free. A classifier learned from initial labeling tends to result in more and

more misclassifications when the data sets expand over time, with more and more unseen data. Therefore, it is worth considering how to incorporate human guidance further to achieve better results—in other words, how to propagate the human corrections to unlabeled samples in order to fix analogous errors. However, it is expensive to rebuild a classifier from scratch using newly collected training data because it requires a lot of human annotations. It would be nice to reuse the previous labeled data and then incorporate subsequent human verifications or corrections.

Once the reliability of segmentation is below a given threshold or obvious erroneous segmentations are detected, the system queries for active verification actively. Then, the user will examine the results, and correct misclassifications by clicking the corresponding regions with a mouse. Human correction will be propagated to the unlabeled samples to fix analogous misclassifications. The simplest way to conduct this task would be to rebuild or modify the affinity matrix in Eq. (2.59) and reperform label propagation. However, this scheme is too inefficient for the verification to be interactive because the inverse of the Laplacian matrix in Eq. (2.61) needs to be recalculated.

Instead, we propose a verification propagation scheme by extending the augmented graph method introduced by Zhu et al. [12] to handle a batch of samples at each round for more effective and efficient interaction. We build an augmented graph by adding auxiliary nodes, which are called *virtual supervisors* and denoted by $\mathcal{S} = \{\mathbf{y}_s\}_{s=1}^{N_s}$, with N_s being the number of virtual supervisors and $\mathbf{y}_s$ being the human label given on the s th sample, and connect them to the corrected samples with weight w , such that $w \rightarrow +\infty$, as shown in Fig. 2.16. Then the adjacency matrix of the augmented graph can be built by adding rows and columns to the original adjacency matrix as follows:

$$\mathbf{W}^+ \leftarrow \begin{bmatrix} \mathbf{W}_{ll} & \mathbf{W}_{lu} & \mathbf{O}_{ls} \\ \mathbf{W}_{ul} & \mathbf{W}_{uu} & \mathbf{W}_{us} \\ \mathbf{O}_{sl} & \mathbf{W}_{su} & \mathbf{O}_{ss} \end{bmatrix} \quad (2.62)$$

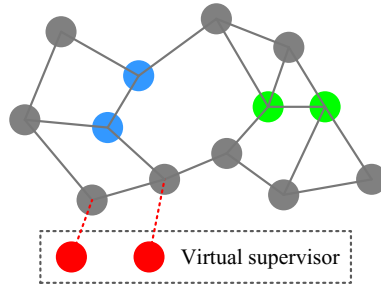


Fig. 2.16 An augmented graph example where *dark gray* and *light gray* nodes indicate differently labeled samples, and *gray* nodes are unlabeled samples. Human labels are given on the most confusing samples by adding auxiliary nodes named *virtual supervisors* and connecting them to the samples with an infinite weight.

where $\mathbf{O}_{ls}$, $\mathbf{O}_{sl}$, and $\mathbf{O}_{ss}$ are zero submatrices with appropriate sizes, and $\mathbf{W}_{us} = w\mathbf{Z}_{us}$ and $\mathbf{W}_{us} = w\mathbf{Z}_{su} = w\mathbf{Z}_{us}^T$, where $\mathbf{Z}_{us}$ and $\mathbf{Z}_{su}$ are binary indicator matrices indicating which virtual supervisor is connected to which unlabeled sample.

The label propagation over this augmented graph can be obtained by minimizing the following objective function [4]:

$$f(\mathbf{Y}_u) = \text{tr} \left([\mathbf{Y}_l; \mathbf{Y}_u; \mathbf{Y}_s]^T \mathbf{L}^+ \begin{bmatrix} \mathbf{Y}_l \\ \mathbf{Y}_u \\ \mathbf{Y}_s \end{bmatrix} \right) \quad (2.63)$$

where $\mathbf{L}^+$ is the Laplacian matrix of $\mathbf{W}^+$ and $\mathbf{Y}_s$ is the label matrix of virtual supervisors. Minimizing this objective yields a closed-form solution:

$$\begin{aligned} \mathbf{Y}_u^+ &= -(\mathbf{L}_{uu}^+)^{-1} [\mathbf{L}_{ul} \mathbf{L}_{us}] \begin{bmatrix} \mathbf{Y}_l \\ \mathbf{Y}_s \end{bmatrix} \\ &= -\mathbf{\Gamma}_{uu}^+ (\mathbf{L}_{ul} \mathbf{Y}_l - w\mathbf{Z}_{us} \mathbf{Y}_s) \end{aligned} \quad (2.64)$$

where $\mathbf{L}_{uu}^+$ denotes the Laplacian submatrix of the augmented graph corresponding to the unlabeled samples; $\mathbf{\Gamma}_{uu}^+$ is its inverse; and $\mathbf{L}_{us} = -\mathbf{W}_{us} = -w\mathbf{Z}_{us}$.

In the rest of this section, we will show how to efficiently compute $\mathbf{\Gamma}_{uu}^+$ and $\mathbf{Y}_u^+$. The diagonal submatrix of the augmented graph corresponding to the unlabeled samples $\mathbf{D}_{uu}^+$ is computed as

$$\begin{aligned} \mathbf{D}_{uu}^+(i, i) &= \sum_{j=1}^{N_l} \mathbf{W}_{ul}(i, j) + \sum_{j=1}^{N_u} \mathbf{W}_{uu}(i, j) + \sum_{j=1}^{N_s} \mathbf{W}_{us}(i, j) \\ &= \mathbf{D}_{uu}(i, i) + w(\mathbf{Z}_{us} \mathbf{Z}_{su})(i, i) \end{aligned} \quad (2.65)$$

We can easily prove that $\sum_{j=1}^{N_s} \mathbf{Z}_{us}(i, j) = (\mathbf{Z}_{us} \mathbf{Z}_{su})(i, i)$ with the fact that $\mathbf{Z}_{us}$ is a binary indicator matrix. Note that $\mathbf{Z}_{us} \mathbf{Z}_{su}$ is a diagonal matrix. Then, $\mathbf{L}_{uu}^+$ can be computed as

$$\begin{aligned} \mathbf{L}_{uu}^+ &= \mathbf{D}_{uu}^+ - \mathbf{W}_{uu}^+ \\ &= \mathbf{L}_{uu} + w\mathbf{Z}_{us} \mathbf{Z}_{su} \end{aligned} \quad (2.66)$$

and accordingly, $\mathbf{\Gamma}_{uu}^+$ can be calculated using the binomial inverse theorem:

$$\begin{aligned} \mathbf{\Gamma}_{uu}^+ &= (\mathbf{L}_{uu}^+)^{-1} \\ &= (\mathbf{L}_{uu} + w\mathbf{Z}_{us} \mathbf{Z}_{su})^{-1} \\ &= \mathbf{L}_{uu}^{-1} - w\mathbf{L}_{uu}^{-1} \mathbf{Z}_{us} (\mathbf{I}_{N_s} + w\mathbf{Z}_{su} \mathbf{L}_{uu}^{-1} \mathbf{Z}_{us})^{-1} \mathbf{Z}_{su} \mathbf{L}_{uu}^{-1} \\ &= \mathbf{\Gamma}_{uu} - \mathbf{\Gamma}_{uk} (\mathbf{I}_{N_s} / w + \mathbf{\Gamma}_{kk})^{-1} \mathbf{\Gamma}_{ku} \end{aligned} \quad (2.67)$$

where $\mathbf{I}_{N_s}$ is an $N_s \times N_s$ identity matrix and $\mathbf{\Gamma}_{uk}$, $\mathbf{\Gamma}_{kk}$, and $\mathbf{\Gamma}_{ku}$ are submatrices of $\mathbf{\Gamma}_{uu}$ such that $\mathbf{\Gamma}_{uk} = \mathbf{\Gamma}_{uu} \mathbf{Z}_{us} = [\mathbf{\Gamma}_{uu}(i, j)]$, $\forall i \in \mathcal{U}, j \in \mathcal{K}$; $\mathbf{\Gamma}_{kk} = [\mathbf{\Gamma}_{uu}(i, j)]$, $\forall i \in \mathcal{K}, j \in \mathcal{K}$; and $\mathbf{\Gamma}_{ku} = \mathbf{Z}_{su} \mathbf{\Gamma}_{uu} = [\mathbf{\Gamma}_{uu}(i, j)]$, $\forall i \in \mathcal{K}, j \in \mathcal{U}$, where $\mathcal{K}$ is a set of selected samples among $\mathcal{U}$ for human verification.

By substituting Eq. (2.67) into Eq. (2.64), we obtain

$$\begin{aligned}\mathbf{Y}_u^+ &= -\mathbf{\Gamma}_{uu}^+(\mathbf{L}_{ul}\mathbf{Y}_l - w\mathbf{Z}_{us}\mathbf{Y}_s) \\ &= -(\mathbf{\Gamma}_{uu} - \mathbf{\Gamma}_{uk}(\mathbf{I}_{N_s}/w + \mathbf{\Gamma}_{kk})^{-1}\mathbf{\Gamma}_{ku})\mathbf{L}_{ul}\mathbf{Y}_l + w(\mathbf{\Gamma}_{uu} - \mathbf{\Gamma}_{uk}(\mathbf{I}_{N_s}/w + \mathbf{\Gamma}_{kk})^{-1}\mathbf{\Gamma}_{ku})\mathbf{Z}_{us}\mathbf{Y}_s\end{aligned}\quad (2.68)$$

Let $w \rightarrow +\infty$; then, we can obtain

$$\begin{aligned}\mathbf{Y}_u^+ &= \mathbf{Y}_u - \mathbf{\Gamma}_{uk}\mathbf{\Gamma}_{kk}^{-1}\mathbf{Y}_k + \mathbf{\Gamma}_{uk}\mathbf{\Gamma}_{kk}^{-1}\mathbf{Y}_s \\ &= \mathbf{Y}_u + \mathbf{\Gamma}_{uk}\mathbf{\Gamma}_{kk}^{-1}(\mathbf{Y}_s - \mathbf{Y}_k)\end{aligned}\quad (2.69)$$

where $\mathbf{Y}_k$ is a submatrix of $\mathbf{Y}_u$ that is constructed by stacking the rows of $\mathbf{Y}_u$ that correspond to queried samples.

We denote $\mathbf{\Gamma}_{uk}\mathbf{\Gamma}_{kk}^{-1}$ as the *verification propagation matrix*. By propagating the human verification information through this matrix, the segmentation results can be incrementally improved. Note that we compute the inverse of $\mathbf{\Gamma}_{kk}$, which is an $N_s \times N_s$ matrix; rather than $\mathbf{L}_{uu}^+$, which is an $N_u \times N_u$ matrix. As a result, verification propagation can be efficiently performed, leading to efficient interactive segmentation.

Examples of the cell segmentation results for the subsequential frames are shown in Fig. 2.17. In Fig. 2.17C and D, we demonstrate the soft and hard segmentation results based on label propagation of the initial human annotation from the first image. As can be

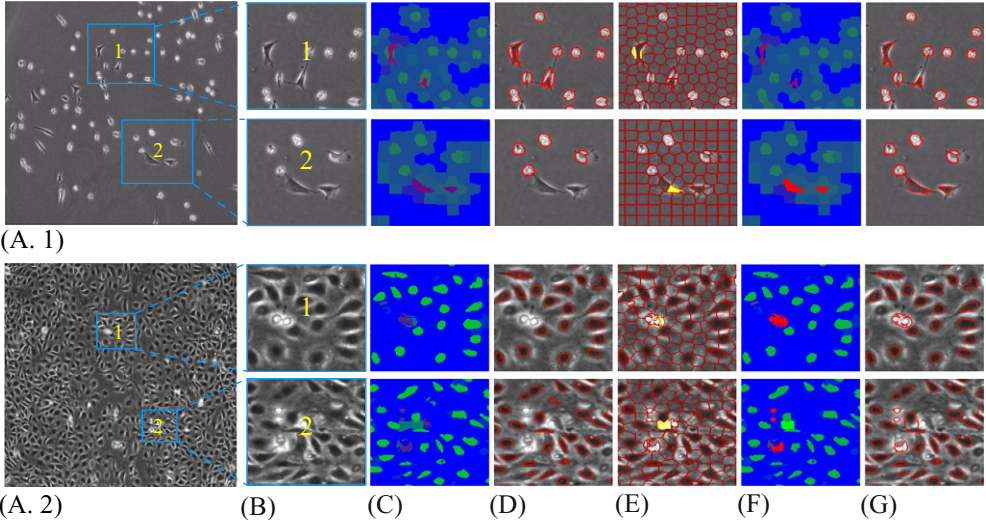


Fig. 2.17 Erroneous segmentation fixing examples with verification propagation on three images from Seq1 (top), Seq2 (middle), and Seq3 (bottom). (A) Input phase contrast microscopy images. (B) Zoom-in subimages. (C) Soft segmentation based on label propagation using annotation acquired in the first frame. (D) Hard segmentation based on the corresponding soft segmentation. (E) Actively selected superpixels for verification marked by *white*. (F) Improved soft segmentation after verification propagation. (G) Improved hard segmentation based on the improved soft segmentation.

seen in Fig. 2.17D, the segmentation result contains several errors (i.e., some dark cells are missed and bright halos are misclassified into cells). These errors are ascribed to inadequate or unbalanced human labels because the phase retardation features of some cells are different from the first frame in the sequence. Some of the superpixels containing errors were automatically selected by our algorithm to be verified and corrected by humans, as is shown in Fig. 2.17E. In Fig. 2.17F and G, we demonstrate the improved soft and hard segmentation results after human correction and its propagation. As shown in Fig. 2.17G, in addition to the errors of superpixels selected by superpixel selection, similar errors of other superpixels are effectively fixed as well.

5. Conclusion

As the amount of data increases in the long-term, high-throughput biological experiments, cell detection and segmentation are recognized as fundamental components in both fundamental biology research and clinical applications. In this chapter, we have presented numerous works on microscopy image analysis by considering the particular imaging mechanism. We introduced work on the understanding of phase contrast microscopy imaging and differential interference contrast microscopy, and then presented the imaging models based on the optics of microscopes. Specifically, the linear imaging models can approximate the imaging formation process, which removes artifacts and segments cells with effortless thresholding. The dictionary-based restoration further boosts the performance by considering the diffraction patterns caused by specimens. Phase retardation features of each cell can be effectively restored by solving a quadratic cost function with a sparse regularization.

With the microscopy image properly restored, we further proposed to segment cells out of the background with a few human annotations. The microscopy images can be partitioned into several superpixels by clustering the neighboring pixels with similar phase retardation features. Cell segmentation is consequently performed by grouping neighboring superpixels with similar identities. If the segmentation result is not reliable, we seek to select a batch of error-prone superpixels to query for verification. Once errors are detected and corrected, the information is systematically propagated over an augmented affinity graph to the unlabeled superpixels. This step is efficiently conducted by introducing a verification correction propagation matrix rather than reperforming the label propagation from the beginning.

Nowadays, deep learning has dramatically improved the state-of-the-art performance of computational models in many domains by learning representations of data with multiple levels of abstraction. The large volumes of image data captured from the long-term, high-throughput biological experiments make it possible to develop a deep architecture model. It is believable that a deep neural network such as CNNs would promote the progress of microscopy image analysis. Nevertheless, the performance of CNN would be improved if we could take the particular imaging mechanism into account.

Acknowledgments

National Key Research and Development Program of China (No.2020AAA0104304), NSFC Project 62076147. This project was supported by NSF CAREER Award IIS-2019967.

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CHAPTER 3

Detection and segmentation in microscopy images

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1. Introduction

Detection and segmentation of objects of interest play a critical role in the analysis of anatomical structures for various biomedical applications. Detection is generally considered to be a precursor for solving higher-level problems, such as segmentation, tracking, and morphological analysis. There are multiple challenges seen in microscopy images for detection of objects of interest, such as low signal-to-noise ratio, low contrast, overlapping objects, low resolution, and variation in appearance. A significant challenge in biomedical image segmentation is the variability in the appearance of objects, such as contrast and texture, due to different imaging conditions and settings. Typical approaches used for detection and segmentation are based on one of a number of categories, such as feature thresholding, region-based clustering, deformable models, and supervised learning approaches. The various supervised learning techniques that we focus on include hierarchical methods and deep learning approaches using convolutional neural networks (CNNs) for detection and segmentation in optical and electron microscopy images.

Hierarchical methods encompass segmentations of an image at various detail levels and thresholds, the higher-level regions can be obtained from a combination of regions from segmentations at lower levels [5, 10, 26, 27, 30, 51, 52, 79, 90]. Hierarchical methods have been used to solve both detection and segmentation problems in microscopy images. Hierarchical methods preserve the spatial and neighboring information among segmented regions. A number of methods, such as the watershed transform [14], an ultrametric contour map (UCM) [8], Felzenszwalb's method [25], and an MSER detector [55], can be used to generate superpixels in hierarchical approaches. *Superpixels* are a group of pixels in spatial proximity that have similar intensity and texture, thereby preserving the boundaries' features [1, 81].

CNNs provide state-of-the-art solutions for various computer vision applications [31, 42, 53, 65] and have been used extensively to solve various classification problems. However, in many biomedical imaging applications, the desired output requires pixelwise predictions (i.e., a class label is predicted for each pixel). A patch-based approach was introduced to address problems such as segmentation of neural membranes and mitosis detection using CNNs, where the pixels are classified based on the neighborhood region [19, 20]. The patch-based classification relies on the size of the patch to determine the localization accuracy and contextual information. Even though CNNs have been the method of choice for object recognition and classification tasks, their performance on dense output prediction problems for biomedical applications needs to be improved. Recently, fully convolutional networks (FCNs) that can be trained in an end-to-end manner have been used for semantic labeling tasks, where a class label is assigned to each pixel [53]. Based on the FCN, U-Net architecture was introduced for biomedical image segmentation. U-Net architecture has been the preferred network for various applications in the field of medical imaging. In Fakhry et al. [24], residual connections were introduced between the encoder and decoder parts of a network to segment neurons in electron microscopy images. CNNs and FCNs have also been used to detect cell centers [88] and count cells by predicting cell density [84] in microscopy images.

There are two significant challenges that limit the performance of CNNs in biomedical applications: (1) large-labeled training data sets with thousands of training images that are well distributed to capture the potential variability of the data, (2) and the training data distribution needs to be representative of the images that the model is used to make predictions for. It is tedious and expensive to acquire fine-grained, pixel-level manual annotations to learn models to predict segmentation labels. There may be discrepancies between the training and testing distributions when the learned model is deployed to make predictions for images collected with different settings, illumination, contrast, and varying image acquisition techniques. In this chapter, we focus on two distinct applications in biomedical imaging: (1) detection and segmentation of cells in optical microscopy images; and (2) and segmentation of neuronal structures in electron microscopy images.

1.1 Cell detection and segmentation in optical microscopy images

A cell is considered to be the fundamental, structural, and functional building unit of all living organisms. Cells have the ability to divide and replicate and are responsible for the evolution of an organism. Studying the underlying principles related to cell theory is critical to help gain perspective on cellular mechanics. Due to the advancement in the field of microscopic imaging techniques, it is feasible to acquire very large numbers of images in a short span of time. The abundance and complexity in the data generated using modern equipment drives us to pursue computerized techniques for cellular image analysis. Automated image segmentation for cell analysis is a challenging problem due to the large variability (different microscopes, stains, cell types, and cell densities) and complexity of the data (time-lapse, large numbers of cells). Accurately detecting a large number of cells in dense images is difficult due to multiple problems, such as occlusion from overlapping of several objects in the depth direction and spatially overlapping cells. Therefore, it is important to develop robust generic algorithms that can be used on a variety of cells from different modalities. Next, we discuss the current contributions in the field of cellular image analysis and extrapolate possible improvements.

1.1.1 Intensity thresholding and filtering

Intensity thresholding based on the distribution of pixel values is considered to be one of the earliest approaches used for cell segmentation [56, 83]. A global or adaptive threshold is chosen to distinguish cell regions from its background, which is feasible only if the cells have homogeneous intensities, have a high contrast with background, and are distinctly separable. Generally, intensity thresholding is used as a preliminary step in algorithms to eliminate background clutter. The intensity images can also be transformed to represent some derived features such as edges and blobs using filtering operations. Edge detection is used to delineate cell contours [17, 54] and is often used in the initial stages of various segmentation algorithms as a feature detector [4, 16]. Morphological filtering is used frequently as a preprocessing or postprocessing step in cell detection and segmentation [7, 22, 62]. Detection and segmentation based on local image information can be erroneous because they can be associated with imaging artifacts or noise.

1.1.2 Region-based segmentation

Region-based segmentation, in which pixels that belong to a cell are grouped, is a popular approach for cell segmentation. Typically, region-growing algorithms start from selected seed points and grow regions by augmenting more connected points that have properties similar to the seed points, such as specific ranges of intensity or color, to form a labeled image [2]. Region-based segmentation approaches also use clustering algorithms with no seed points, where similar image pixels are aggregated in an unsupervised

framework. Gaussian mixture models were used to cluster pixels using color and texture features [11]. The watershed transform is one of the more popular region accumulation methods; it requires an edge-enhanced image to segment an image into multiple regions and catchment basins. In most scenarios, this method produces oversegmentation and requires further processing. Hence, several algorithms use the watershed transform as an initial step in their segmentation pipeline.

Hierarchical image segmentation has been demonstrated to be an effective method for cell detection and segmentation [5, 10, 26, 27, 30, 51, 52, 78, 79, 90]. A hierarchical representation for the segmentation is constructed using the cell candidates, such that higher-level regions are built by merging regions at lower levels. Supervised algorithms with handcrafted features are used to train classifiers to merge the regions in the hierarchical representation [26, 28, 50, 51, 76]. Alternatively, learning algorithms can also be used to predict probability scores for the cell candidates [26, 64, 79]. Dynamic programming or integer linear programming is used to find a nonoverlapping set of candidates from a hierarchical representation. It is challenging to choose a good feature set to discriminate target cells from the background because feature selection relies on prior knowledge about the data.

Images can also be represented using a graphical representation, where pixels are the nodes and the edge weights are based on the similarity with neighbors [70, 92]. The segmentation problem is then solved as a graph-partitioning problem. To reduce the computational load, the graphical representation has also been constructed using superpixels. In Zhang et al. [92], a region adjacency graph is constructed using superpixels, and then the segmentation is inferred by partitioning the graph using correlation clustering.

1.1.3 Deformable models

Cell contours can be represented using deformable models. Among these methods, variational image segmentation with level sets has been a well-known choice due to attractive properties such as adaptive topology, which can naturally change during evolution [15, 61]. Level sets evolve by minimizing an energy functional that is chosen to distinguish the foreground from the background. An application of level sets to segment cells is seen in Nath et al. [58]. However, due to the nonparametric nature of level sets, level-set propagation always has to include a regularization term, such as a penalty on curve length, surface area, or curvature [21]. Implicit parametric shape models (namely, disjunctive normal shape models) are used to accurately learn cell boundaries in Ramesh et al. [63]. Implicit parametric models have the advantage of being able to easily learn statistics and naturally change topology during its evolution. Evolving the boundaries for individual cells can get computationally expensive because the time complexity increases with the density of cells in the images.

1.1.4 Convolutional neural networks

CNNs have been used recently for semantic labeling tasks, where a class label is assigned to each pixel. A FCN [53] was proposed for semantic segmentation that can be trained in an end-to-end manner. FCNs avoid redundant computations; however, this comes at the cost of localization accuracy. Based on the FCN, U-Net architecture was introduced to segment cells in biomedical images [66]. The architecture uses skip connections between the encoder and decoder parts of the network to enable precise localization. The U-Net architecture relies on strong data augmentation strategies. It is difficult to obtain pixel-level annotations to train a CNN to segment cells in microscopy images, and training a network with limited labeled data makes the task challenging. To enhance the performance of CNNs, specialized feature maps like eigenvalue of Hessian and wavelet-filtered images have been used with raw intensity images for training [67]. In Akram et al. [3], two networks are used—one for predicting cell proposals trained using the ground truth bounding boxes, and the other for generating cell segmentation masks for the proposed candidates trained using pixel-level labels.

Conventionally, CNNs have been used for solving classification and segmentation problems. CNNs can also be used to solve cell detection problems [48, 84–88]. Cell detection is a critical problem in biomedical image analysis, which is often a prerequisite for segmentation. It is difficult to detect cells when there is a high degree of overlap between them. A good subset of cell proposals generated with different parameters are used to train a CNN to learn to predict its probability of being a cell [48]. In Xie et al. [85], a CNN was modified to represent a regression model on a structured proximity space for cell detection. A FCN-based framework trained end on end on image patches was introduced by Xie et al. [84] for predicting a spatial density map of target cells, and the number of cells is estimated by integration over the learned density map. A similar approach is seen in Xue et al. [88], where the global cell count is cast as a regression problem using CNNs. A CNN-based regression network was used to predict the compressed vectors that encode the output pixel space indicating cell centers from the input pixels [87]. In Sirinukunwattana et al. [71], a spatially constrained CNN (SC-CNN) regresses the likelihood of a pixel being the center of a nucleus to detect nuclei and uses a locality sensitive CNN for classification. A structured regression model based on a fully residual CNN was used for efficient cell detection [86].

1.2 Neuronal segmentation in electron microscopy images

The study of neural circuit reconstruction (i.e., connectomics) remains one of the most challenging problems in neuroscience. Image analysis techniques are used to study the connectivity of nervous systems. Electron microscopy (EM) images have been used for neural circuit reconstruction because they provide high resolution of the intracellular structures,

synapses, and gap junctions. Automated and semiautomated 3D reconstruction of neurites in EM stacks can be tremendously helpful for accelerating connectomics research [9]. The main goal in this problem is to identify neurons that span multiple adjacent image slices and reconstruct them. Conventionally, the 3D reconstruction problem can be broken into two tasks: (1) 2D prediction task, where each image slice is segmented independently; and (2) a postprocessing step on segmented slices to reconstruct the 3D neurites by linking similar segments across the entire image stack. Supervised algorithms with handcrafted features are used to learn classifiers for the postprocessing step to link similar segments together across the entire image stack. The classifiers incorporate prior knowledge and understanding of the data; hence, classifiers trained on a particular data set may not perform well when tested on data sets acquired in a different scenario. Alternatively, there are methods that seek to achieve 2D segmentation and 3D reconstruction at the same time. Next, we discuss the current contributions in the field of neural circuit reconstruction and extrapolate possible improvements.

1.2.1 Traditional filtering approaches

There have been numerous unsupervised algorithms that have been proposed to segment 2D neurons [38, 44, 74]. Anisotropic directional filtering was used to enhance the contrast and continuity of cell membranes [38, 74]. These filtering approaches do not remove intracellular structures, reducing the accuracy for membrane segmentation. In Kumar et al. [44], radon-like features were introduced to aggregate desired information derived from images and suppress undesirable intracellular structures.

1.2.2 Supervised learning

Supervised learning methods have been used extensively for segmenting membranes and neurons in 2D. These include single-layer networks [57], cascades of classifiers [37, 68], and hierarchical contextual models [69]. A single-layer artificial neural network (ANN) with Hessian eigenspace features was used for segmentation [57]. A cascade of ANNs were used [37]. Several ANNs are used in series; each ANN uses the classification context provided by the previous network to improve detection accuracy [37]. The model in Seyedhosseini [68] uses contextual information along with radon-like features to learn about a series of discriminative models. The multiscale model representation of the context image was used to give the subsequent classifiers access to a larger contextual area for better detection of neuron membranes in EM images. In Seyedhosseini and Tasdizen [69], a multiclass, multiscale contextual model was introduced that exploits both input image features and contextual information with a logistic disjunctive normal network classifier to detect membranes. Region-based methods are used to transform the membrane probability maps into cell segments; they are required for improving the segmentation accuracy from membrane detections in EM images [9].

A hierarchical merge tree structure was built to represent multiple region hypotheses, and the merge tree was resolved with consistency constraints to acquire final intrasection segmentation [50]. The sections were linked to reconstruct the intersection neurons using supervised learning. In Hu et al. [34], unsupervised learning and signal-processing techniques are used to obtain high depth-resolution EM images computationally without sacrificing throughput. Several methods have been used to link similar neuron sections across the image stack. A graph-cut framework was proposed to trace 2D contours in 3D [89]. In Vitaladevuni and Basri [82], the neuron segments are jointly clustered based on their likelihood of belonging to the same neuron. The 3D linking was formulated as a quadratic optimization problem by coclustering each pair of adjacent segments. An automatic processing pipeline was introduced that combines the probabilistic output of a random forest classifier with geometrical consistency constraints to optimize the 3D affinity matrix [40]. In Laptev [45], global dense correspondences between the sections was established using a SIFT-Flow [49] algorithm in order to evaluate the features of the corresponding pixels and use them to perform the segmentation.

Alternatively, there are a set of methods that perform 2D segmentation and 3D reconstruction simultaneously. A reinforcement learning framework was proposed to merge supervoxels into segmentations [36]. A supervoxel merging method was proposed for segmenting neurons, all merging decisions were coupled in a graphical model that incorporates both supervoxel face and boundary curve information [6]. The use of supervoxels allows the extraction of expressive geometric features that are used by the higher-order potentials in the graphical model. Multiple 2D segmentation hypotheses were generated for each section in the stack, and segments were fused to provide the most globally consistent segmentation of the stack in a Markov random field framework [80]. In a similar manner, 2D segmentation hypotheses were generated as a tree structure, and then the 2D segmentation and 3D reconstruction were formulated as an integer linear program with constraints [29]. A reinforcement learning framework was used to learn a merging policy for superpixel agglomeration [60]. There are also other semiautomatic methods that utilize user input to improve the segmentation of neurons. Neurons are oversegmented into 2D and 3D regions, and the regions that belong to one neuron are manually merged [18]. Skeleton reconstructions of neurons are combined with automated volume segmentations [13]. To be able to segment image volumes in 3D requires the data to be mostly isotropic. There are limitations to the image volume size that can be generated with isotropic resolution. Direct 3D segmentation approaches may not be feasible for anisotropic data sets.

1.2.3 Convolutional neural networks

Even though deep networks give state-of-the-art solutions for object recognition and classification tasks, their performance on dense output prediction problems has room for improvement, especially in neural circuit reconstruction applications using high-resolution EM images [23, 35, 46, 73, 77]. A convolutional network was trained to

classify pixels as membrane or nonmembrane [35]. Patch-based approaches for dense output prediction problems have the conflicting goals of full-resolution prediction and incorporation of sufficient contextual information [41]. A deconvolution network was introduced [59] for semantic segmentation. Residual deconvolutional networks (RDNs) were proposed to balance the tradeoff between increasing contextual information to exploit the multiscale features and the ability to preserve the pixel-level resolution and accuracy expected for dense output prediction [24]. Multiple residual shortcut paths were added to a fully deconvolutional network with minimum additional computations and the final pixel-level predictions incorporate features from both pathways. To achieve human-level performance, Zeng et al. [91] employ a combination of novel boundary map generation methods, with optimized model ensembles using CNNs to address the challenges of segmenting anisotropic image EM stacks. Their deep network for 3D neurite boundary detection deals with anisotropy and misalignment problems that are commonly seen in 3D EM data. A variant of 3D U-Net architecture was proposed [47] to learn two tasks. The primary task predicts affinities between nearest-neighbor voxels, and the auxiliary task predicts long-range affinities. The learned network is used to generate oversegmented supervoxels, which are greedily agglomerated based on mean affinity. Data augmentation is used to address challenges such as misalignments, missing sections, and out-of-focus sections in EM images.

2. Cell detection and segmentation in optical microscopy images

In this section, we focus on methods for detection and segmentation of cells in optical microscopy images. Because detection is a preliminary step for classification, we also review an approach for nucleus detection and classification.

2.1 Learning to detect cells using nonoverlapping extremal regions

A generic learning-based method was introduced for detecting cells across different microscopy modalities [10]. A highly efficient MSER detector [55] was used to generate cell candidate regions. Let $R_1, R_2, \dots, R_N$ be the set of N cell candidate regions in an image. A feature vector $\mathbf{f}_i$ was computed for every cell candidate region R_i . A classifier with weight vector $\mathbf{w}$ was learned to predict V_i , the probability of the region belonging to the class of cells of interest. The classifier model was learned in a structured SVM framework from dot-annotated training data to score each candidate region for the detection task. A subset of cell candidates was picked such that the sum of scores of the picked regions was maximized while satisfying the nonoverlap constraint (i.e., the chosen regions did not overlap). Binary indicator variables $\mathbf{y} = \{y_1, y_2, \dots, y_N\}$ are used to identify the regions that are picked. Let $\dagger$ be a set of those region subsets that do not overlap; that is, $\dagger = \{\mathbf{y} \mid \forall i, j: (i \neq j) \wedge (y_i = 1) \wedge (y_j = 1) \Rightarrow R_i \cap R_j = \emptyset\}$. The optimization problem was formulated as

$$F(\mathbf{y}) = \max_{\mathbf{y} \in \dagger} \sum_{i=1}^N y_i V_i \quad (3.1)$$

The nestedness property of the extremal regions was exploited to obtain an exact solution of Eq. (3.1) via dynamic programming. They assume a set of M training images $\mathcal{I}^1, \mathcal{I}^2, \dots, \mathcal{I}^M$, where each training image $\mathcal{I}^j$ has a set of N^j MSER regions $(R_1^j, R_2^j, \dots, R_{N^j}^j)$. The training images are annotated such that n_i^j denotes the number of user-placed dots (annotations) inside the region R_i^j . For each of these regions R_i^j , a feature vector $\mathbf{f}_i^j$ was computed. The goal of learning, then, is to find a weight vector $\mathbf{w}$ so that the regions with $n_i^j = 1$ are selected, and also to ensure that for each dot, a region is selected that contains it. A structured SVM was used to optimize the performance of the inference procedure on the training data. The configuration $\mathbf{y}^j \in \dagger^j$ is the set of nonoverlapping regions for the image $\mathcal{I}^j$. The researchers introduced a loss function associated with $\mathbf{y}^j$ to measure the deviation from the one-to-one correspondence between the user-placed dots and the picked regions:

$$L(\mathbf{y}^j) = \sum_{i=1}^{N^j} y_i^j |n_i^j - 1| + U^j(\mathbf{y}^j) \quad (3.2)$$

where $U^j(\mathbf{y}^j)$ denotes the number of user-placed dots that have no correspondences and are not covered by any region R_i^j with $y_i^j = 1$. The ground truth configurations $\bar{\mathbf{y}}^j = \{\bar{y}_1^j, \bar{y}_2^j, \dots, \bar{y}_{N^j}^j\} \in \dagger$ is determined for every training image, such that a unique extremal region is assigned to every dot annotation. The optimal weight vector $\mathbf{w}$ was found by minimizing the following convex objective:

$$L(\mathbf{w}) = \frac{1}{2} \|\mathbf{w}\|^2 + \frac{C}{M} \sum_{j=1}^M \max_{\mathbf{y}^j \in \dagger^j} \left(\sum_{i=1}^{N^j} (\mathbf{w} \cdot \mathbf{f}_i^j) y_i^j - \sum_{i=1}^{N^j} (\mathbf{w} \cdot \mathbf{f}_i^j) \bar{y}_i^j + L(\mathbf{y}^j) \right) \quad (3.3)$$

where the first term is a regularizer on the weights $\mathbf{w}$, C is a scalar regularization parameter, and the maximum inside the sum represents a convex upper bound on the loss defined in Eq. (3.2).

They evaluate their method on three data sets for cell detection: (1) ICPR 2010 histopathology images of stained breast cancer tissue [32], (2) fluorescence microscopy images of human embryonic kidney (HEK) cells [12], and (3) phase-contrast images of cervical cancer cell colonies of the HeLa cell line (Fig. 3.1). Each of the data sets imposes a different challenge for detection. In the histopathology images, it is difficult to identify the lymphocyte nuclei from the breast cancer nuclei due to their similar appearance. The detection task in the HEK cells is challenging due to the significant intensity variations among cells across the image, fading boundaries, and frequent cell clumping. The phase-contrast HeLa cells present a high variability in cell shapes and sizes.

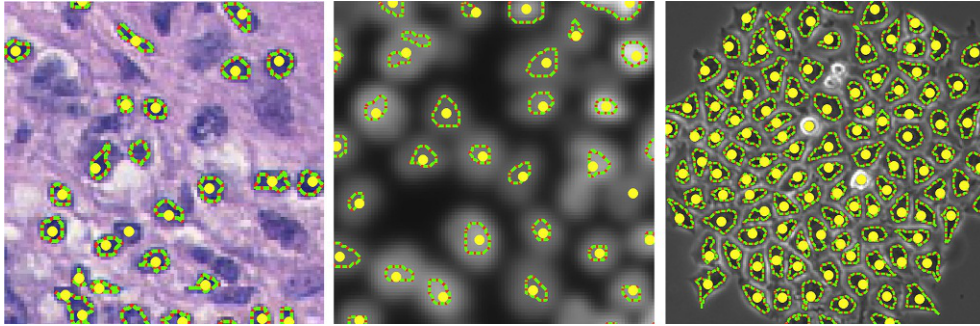


Fig. 3.1 Example images from the data sets used for cell detection. Left: Histopathology image of breast cancer tissue, fluorescence microscopy image of human embryonic kidney cells. Right: Phase-contrast HeLa cells. (C. Arteta, V. Lempitsky, J.A. Noble, A. Zisserman, *Learning to detect cells using non-overlapping extremal regions*, in: *Medical Image Computing and Computer-Assisted Intervention—MICCAI: 15th International Conference, Nice, France, October 1–5, 2012, Proceedings, Part I*, Springer Berlin Heidelberg, 2012, pp. 348–356, Figure 3. Reprinted with permission.)

They evaluate the performance of the algorithm for detecting cell centroids using precision, recall, and F-measure. The precision and recall metrics are indicators for the number of false positives and false negatives, respectively. For the histopathology images [32], two additional metrics were calculated: (1) mean and standard deviation of the Euclidean distance between detected dots and ground truth dots, and (2) absolute difference between the number of cells found and the ground truth number of cells. Three variations of their method was evaluated in their experiments: (1) direct classification (DC) using a learned binary classifier with a $\mathbf{w}$ vector to evaluate all MSERs to choose the region with the highest score in every set of overlapping regions with positive scores; (2) binary SVM + inference (B + I), where the inference is based on the weight vector learned through binary classification; and (3) structured SVM + inference (S + I), which uses inference with the weight vector learned by the structured SVM. Fig. 3.2 shows the precision-recall curves for the various evaluations. From their experiments, it can be observed that enforcing the nonoverlap constraint increases the accuracy when $\mathbf{w}$ is learned within the structured SVM framework. Table 3.1 compares the quantitative experimental results on the histopathology data set. One of the disadvantages is that the research relies heavily on a robust region detector, and thus the application of the results is limited. Also, the researchers use handcrafted features to train the classifier, which may not capture the variability in the classes.

2.2 U-Net: Convolutional networks for biomedical image segmentation

The U-Net architecture [66] is based on the FCN [53]; it encompasses an encoder to capture context and a symmetric decoder to enable precise localization (Fig. 3.3). The encoder is supplemented with successive layers, where pooling operators are replaced

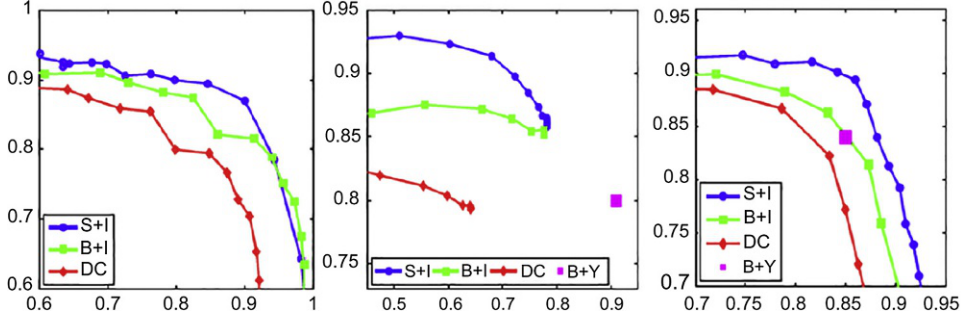


Fig. 3.2 Precision (vertical) versus recall (horizontal) curves for the three data sets for the three variations of their approach and Ref. [12] (denoted as B+Y, where available). Significant improvements brought by the nonoverlap constraint (B+I) and the structured SVM (S+I) can be observed. (C. Arteta, V. Lempitsky, J.A. Noble, A. Zisserman, *Learning to detect cells using non-overlapping extremal regions*, in: *Medical Image Computing and Computer-Assisted Intervention—MICCAI: 15th International Conference, Nice, France, October 1–5, 2012, Proceedings, Part I*, Springer Berlin Heidelberg, 2012, pp. 348–356, Figure 4. Reprinted with permission.)

Table 3.1 Detection results for the ICPR 2010 histopathology images [32].

Method	Precision	Recall	F-score	$\mu_d + \sigma_d$	$\mu_n + \sigma_n$
Arteta et al. [10]	86.99	90.03	88.48	1.68 ± 2.55	2.90 ± 2.13
Bernardis [12]	—	—	—	3.13 ± 3.08	12.7 ± 8.70

by upsampling operators to increase the resolution of the output. For localization, high-resolution features from the encoder are concatenated with the upsampled output. Every layer of the encoder has two 3×3 convolutions, followed by a rectified linear unit (ReLU) and a 2×2 max pooling with stride 2 for downsampling. The number of feature channels is doubled after every downsampling. Similarly, in the decoder, each layer consists of an upsampling of the feature map, followed by an upconvolution of 2×2 and then two 3×3 convolutions with a ReLU layer. The number of feature channels is halved after every upsampling. Skip connections between the encoder and decoder are established by concatenating the layers with the same number of feature channels to introduce high-resolution features to the decoder, enabling precise localization. A pixelwise, softmax layer was used over the final feature map, combined with the cross-entropy loss function to learn segmentation. Thus, the last layer for the segmentation task uses a 1×1 convolution to reduce the number of output channels to the number of labels. The softmax layer is defined as $p_k(x) = \exp(a_k(x)) / (\sum_{k=1}^K \exp(a_k(x)))$, where $a_k(x)$ denotes the activation in feature channel k at the pixel position $x \in \Omega$ with $\Omega \subset \mathbb{Z}^2$ and K is the number of classes. The cross-entropy then penalizes at each position the deviation of $p_{l(x)}$ from 1 using

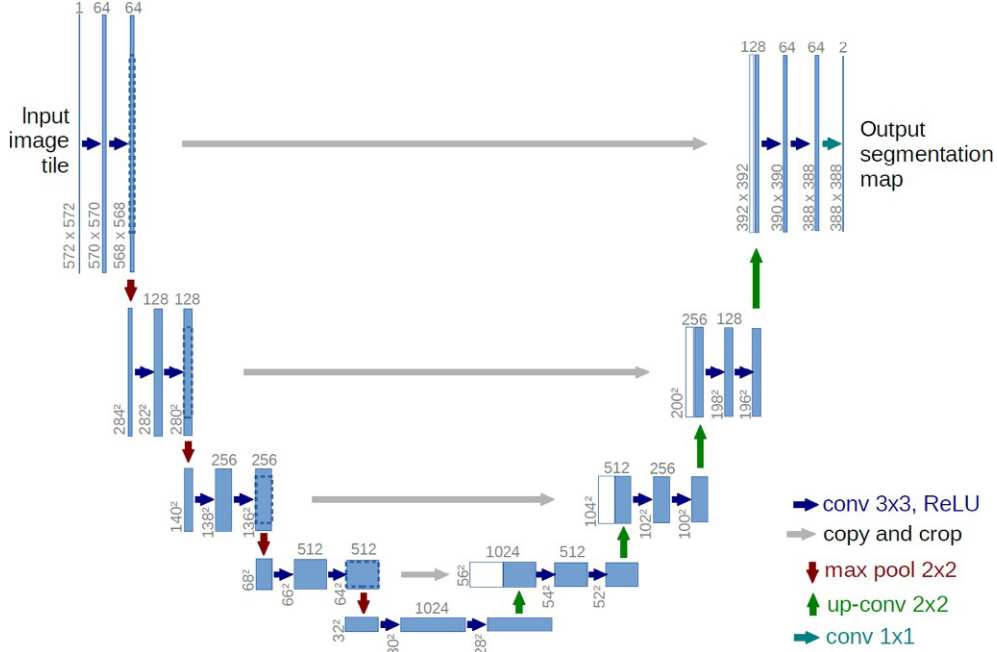


Fig. 3.3 The U-Net architecture (example for 32×32 pixels in the lowest resolution). Each *dark box* corresponds to a multichannel feature map. The number of channels is denoted on top of the box. The x-y-size is provided at the lower-left edge of the box. *White boxes* represent copied feature maps. The *arrows* denote the various operations. (O. Ronneberger, P. Fischer, T. Brox, U-net: convolutional networks for biomedical image segmentation, in: *Medical Image Computing and Computer-Assisted Intervention—MICCAI 2015: 18th International Conference, Munich, Germany, October 5–9, 2015, Proceedings, Part III, 2015, pp. 234–241, Figure 1. Reprinted with permission.*)

$$E = \sum_{x \in \Omega} w(x) \log(p_{l(x)}) \quad (3.4)$$

where $l: \Omega \rightarrow 1, \dots, K$ is the true label of each pixel and $w: \Omega \rightarrow R$ is a weight map used to give importance to the edge pixels. They address the challenge of separating touching cells by using a weighted loss function such that the separating background labels between the touching cells take on a large weight in the loss function. The weight map is precomputed; it forces the network to learn the small separation borders that is introduced between touching cells using morphological operations. The weight map was computed as

$$w(x) = w_c(x) + w_0 \cdot \exp\left(-\frac{(d_1(x) + d_2(x))^2}{2\sigma^2}\right) \quad (3.5)$$

where $w_c: \Omega \rightarrow R$ is the weight map to balance the class frequencies, $d_1: \Omega \rightarrow R$ denotes the distance to the border of the nearest cell, and $d_2: \Omega \rightarrow R$ is the distance to the border of

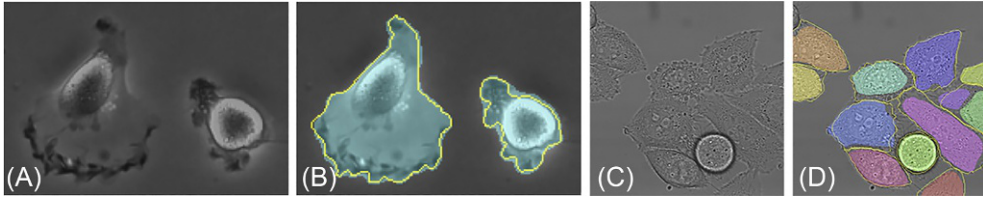


Fig. 3.4 The results of the ISBI cell-tracking challenge. (A) Part of an input image of the PhC-U373 data set. (B) Segmentation result (*highlighted region*) with manual ground truth (*region boundary*). (C) Input image of the DIC-HeLa data set. (D) Segmentation result (*highlighted regions*) with manual ground truth (*yellow border*). (O. Ronneberger, P. Fischer, T. Brox, *U-net: convolutional networks for biomedical image segmentation*, in: *Medical Image Computing and Computer-Assisted Intervention—MICCAI 2015: 18th International Conference, Munich, Germany, October 5–9, 2015, Proceedings, Part III, 2015*, pp. 234–241, Figure 4. Reprinted with permission.)

the second-nearest cell. Data augmentation was used to compensate for the limited training data in biomedical segmentation problems. The network learns invariances to elastic deformations, without the need to see the transformations in the annotated images. The U-Net architecture can be trained in an end-to-end manner.

The U-Net architecture was evaluated for segmentation of neuronal membranes from the EM segmentation challenge [9] and cell segmentation from the ISBI cell-tracking challenge [72] in light microscopic images. The segmentations were evaluated for the PhC-U373 and DIC-HeLa data sets from the cell-tracking challenge. Visual results for the cell segmentation is shown in Fig. 3.4. The U-Net architecture has been used for several biomedical segmentation applications; it requires very few annotated images because it incorporates data augmentation with elastic deformations. The U-Net architecture achieves a warping error of 0.0003529 and a rand error of 0.0382 for the EM segmentation challenge [9], and these results are significantly better than the sliding-window convolutional neural network approach [19], which has a warping error of 0.000420 and a rand error of 0.0504. The U-Net architecture relies on strong data augmentation strategies. Also, the architecture is not robust to variability in the appearance of objects such as contrast and texture due to different imaging settings.

2.3 Improving the robustness of convolutional networks to appearance variability in biomedical images

A neighborhood similarity layer (NSL) [75] was introduced as an additional layer in CNNs to overcome problems posed by variability due to changes in the appearance of objects such as contrast and texture. Data augmentation methods cannot adapt to variations that are not well represented in the training data, whereas the NSL can adapt to unforeseen appearance changes that are not present in the training data set. NSL computes the normalized inner products of feature vectors of the previous layer between a

central pixel that acts as a frame of reference and the spatial neighborhood of that pixel. The NSL is an appearance invariance-inducing transformation on image feature maps. Given an n -dimensional feature map $\phi: \Omega \rightarrow \mathbb{R}^n$ and a neighborhood structure $\mathcal{N}$, the NSL $\psi: \Omega \rightarrow \mathbb{R}^m$ is the vector map of normalized inner products:

$$\psi(\mathbf{x}, \phi, \mathcal{N}) = \left[\frac{\langle \phi(\mathbf{x} + \mathbf{v}) - \bar{\phi}, \phi(\mathbf{x}) - \bar{\phi} \rangle}{(\langle \phi(\mathbf{x} + \mathbf{v}) - \bar{\phi}, \phi(\mathbf{x} + \mathbf{v}) - \bar{\phi} \rangle \langle \phi(\mathbf{x}) - \bar{\phi}, \phi(\mathbf{x}) - \bar{\phi} \rangle)^{1/2}} \right]_{\mathbf{v} \in \mathcal{N}} \quad (3.6)$$

where $\bar{\phi} = (1/|\Omega|) \sum_{\mathbf{x} \in \Omega} \phi(\mathbf{x})$ is the mean feature vector over Ω .

These are some of the favorable properties of the NSL: (1) it is a parameter-free layer, and hence independent of training data; (2) it is not a convolution, but a spatially varying operation that uses the feature vectors at each pixel as a frame of reference; (3) the output feature map ψ has the same dimensionality as the number of pixels in $\mathcal{N}$ and (4) when $\mathcal{N}$ is a square patch, the vector $\psi(\mathbf{x})$ corresponds to a square patch of similarities around and excluding $\mathbf{x}$.

An NSL can be placed after any feature map-producing layer in a network, as shown in Fig. 3.5. Note that a convolutional layer following an NSL operates on a map of feature vectors that correspond to square patches of similarity vectors.

NSL improves the accuracy of regular training and domain adaptation scenarios for cell detection and eye vasculature segmentation. The Fluo-N2DH-GOWT1 and Fluo-N2DL-HeLa data sets used in the experiments are from the cell-tracking challenge [72]. The detection performance was evaluated using the U-Net architecture as a baseline [66] and a modified U-Net architecture with NSL ($9 \times 9 \mathcal{N}$) after the first convolution layer. Cells were detected as locations of the local maxima of the output of the U-Net. Visual results for cell detection on both data sets are displayed in Fig. 3.6. With the addition of

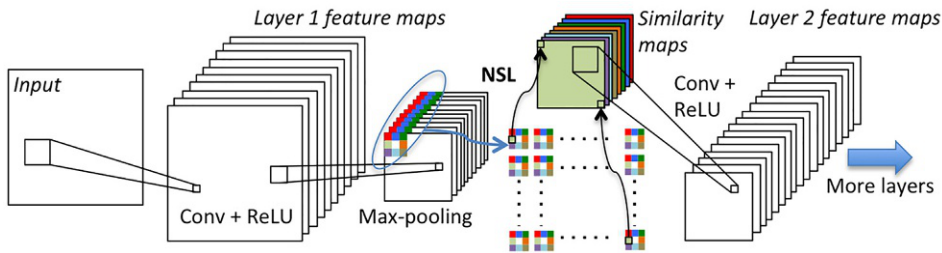


Fig. 3.5 An NSL placed between two layers of a network. At any location $\mathbf{x}$ (the top-left corner is shown), the NSL computes a neighborhood of similarities to $\phi(\mathbf{x})$. This creates an image of two-dimensional neighborhood similarities that are flattened into one-dimensional vectors, creating the output of the NSL. The relationship between a $3 \times 3 \mathcal{N}$ and eight similarity maps are shown with the color coding. (T. Tasdizen, M. Sajjadi, M. Javanmardi, N. Ramesh, *Improving the robustness of convolutional networks to appearance variability in biomedical images*, in: 2018 IEEE 15th International Symposium on Biomedical Imaging (ISBI 2018), 2018, pp. 549–553, Figure 1. Reprinted with permission.)

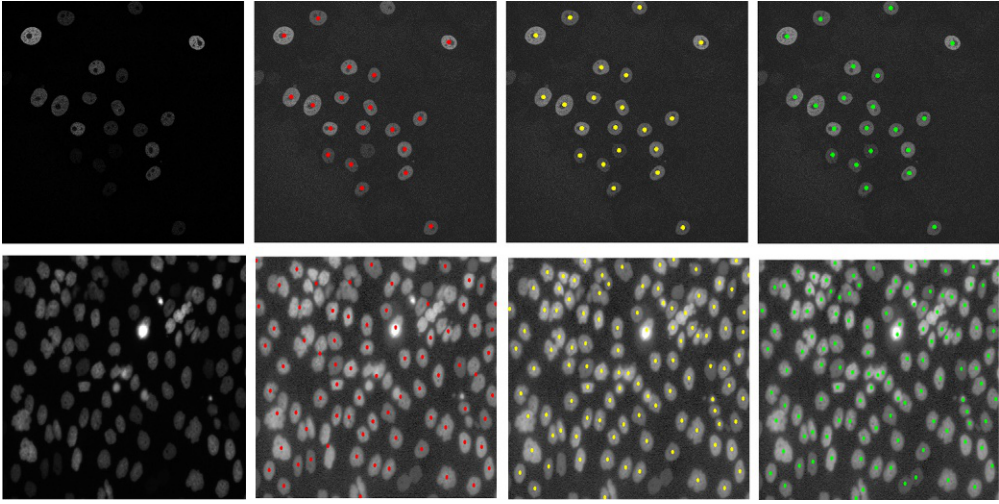


Fig. 3.6 Top: Fluo-N2DH-GOWT1-1. Bottom: Fluo-N2DL-HeLa-2 [72]. Left to right: Test image, predicted centroids without and with NSL, and ground-truth centroids overlaid onto contrast-enhanced images (for display only). (T. Tasdizen, M. Sajjadi, M. Javanmardi, N. Ramesh, *Improving the robustness of convolutional networks to appearance variability in biomedical images*, in: *2018 IEEE 15th International Symposium on Biomedical Imaging (ISBI 2018)*, 2018, pp. 549–553, Figure 2. Reprinted with permission.)

NSL, low contrast cells in the Fluo-N2DH-GOWT1 data set is identified better. In the case of the Fluo-N2DL-HeLa data set, clumped cells are individually identified with the NSL. For quantitative evaluation, the Hungarian algorithm [43] was used to match ground-truth dot annotations with local maxima of the network output. Precision, recall, and F-score were the metrics used to evaluate the detection performance. All the experiments used a single NSL after the first convolutional layer of networks as an appearance invariance-inducing transformation. The detection performance was also compared with that found in Türetken et al. [78] and the U-Net architecture without NSL using contrast-enhanced images. The quantitative results from the evaluations are reported in Table 3.2. The improvement with NSL significantly exceeds the improvement obtained with contrast enhancement of the input images in terms of F-score. The experiments do not use the target data for learning, which facilitates the use of pretrained models with NSL without retraining for domain adaption scenarios. Inclusion of NSL is helpful for discarding the variability seen in cell images.

2.4 Locality sensitive deep learning for detection and classification of nuclei in routine colon cancer histology images

A spatially constrained convolutional neural network (SC-CNN) was introduced to detect nuclei in routine hematoxylin and eosin-stained histopathology images of colorectal

Table 3.2 Cell-detection testing precision, recall, and F-scores.

	Precision	Recall	F-score
Data set	Fluo-N2DH-GOWT1-1		
ECLIP [78]	100.0	31.0	47.33
U-Net	94.91	82.71	88.39
U-Net + contrast-enhanced	94.66	86.16	90.21
U-Net + 9×9 NSL	96.95	94.59	95.75
Data set	Fluo-N2DL-HeLa-2		
ECLIP [78]	78.0	86.0	81.80
U-Net	85.74	69.94	77.03
U-Net + contrast-enhanced	85.86	71.97	78.30
U-Net + 9×9 NSL	89.60	79.49	84.24

Source: T. Tasdizen, M. Sajjadi, M. Javanmardi, N. Ramesh, Improving the robustness of convolutional networks to appearance variability in biomedical images, in: 2018 IEEE 15th International Symposium on Biomedical Imaging (ISBI 2018), 2018, pp. 549–553, Table 1. Reprinted with permission.

adenocarcinoma [71]. The SC-CNN was trained as a spatial regression network that reduces the likelihood of a pixel being the nucleus center. The network utilizes the distance from the center of an object (the nucleus, in this case) to calculate a probability map for detecting that object. SC-CNN contains a layer that was specifically designed to predict the centroid locations of nuclei, as well as the confidence of the locations whether or not they correspond to true centroids. The spatial constraints were used to assign high probability values to pixels in the vicinity of the centers of nuclei. A neighboring ensemble predictor (NEP) was used with the SC-CNN to predict the class label of the detected cell nuclei. A weighted ensemble of local predictors leverages all relevant patch-based predictions in the local neighborhood of the nucleus to yield an accurate classification. Their approach does not require segmentation of the nuclei; rather, they use a sliding window to train the networks on small patches instead of the whole image.

The network is trained on image patches and probability maps representing the nuclei centers. The probability map that is defined has a high peak in the vicinity of the center of each nucleus and is flat elsewhere. The architecture used in SC-CNN is inspired by the results of Ciresan et al. and Krizhevsky et al. [19, 42]. The network consists of conventional layers such as input, convolution, spatial max-pooling, and fully connected layers. The last three layers of the SC-CNN are a fully connected layer F, a parameter estimation layer S1, and a spatially constrained layer S2. Parameters such as the center of the nucleus and the height of the probability mask are estimated in the S1 layer. The predicted output probability is generated from the spatially constrained layer S2 of the network using the estimated parameters for the nucleus. An example of the training patch, the training output patch, and the last three layers of the SC-CNN is shown in Fig. 3.7. The weight vectors and biases was learned by solving the following loss function: